Unsupervised Machine Learning

UNIT – I: Introduction to Unsupervised Learning

1. Define Unsupervised Learning. How is it different from Supervised Learning?
2. What are the major applications of Unsupervised Learning?
3. Explain the general workflow of unsupervised learning.
4. What are the challenges in unsupervised learning?
5. Compare clustering and dimensionality reduction.

UNIT – II: Clustering Techniques – I (K-Means)

1. What is clustering? Explain the K-Means algorithm with an example.
2. How do you choose the value of K in K-Means?
3. What are the limitations of K-Means clustering?
4. What is the Elbow method? How is it used?
5. Explain the steps involved in implementing K-Means in Python.

UNIT – III: Clustering Techniques – II (Hierarchical & DBSCAN)

1. Explain Hierarchical clustering and its types (Agglomerative & Divisive).
2. What is a dendrogram? How is it used in hierarchical clustering?
3. Compare K-Means and Hierarchical Clustering.
4. Explain DBSCAN algorithm with example.
5. What are the advantages of DBSCAN over K-Means?

UNIT – IV: Dimensionality Reduction

1. What is dimensionality reduction? Why is it important?
2. Explain Principal Component Analysis (PCA) with a step-by-step example.
3. Compare PCA and t-SNE.
4. What are eigenvalues and eigenvectors in PCA?
5. Explain feature selection vs feature extraction.

UNIT – V: Evaluation & Applications of Clustering

1. How do you evaluate the performance of a clustering algorithm?
2. Explain the Silhouette score and its interpretation.
3. What is intra-cluster and inter-cluster distance?
4. List and explain real-world applications of clustering.
5. Discuss the role of unsupervised learning in recommendation systems